

KELLY ANN HEILMAN

Laboratory of Tree Ring Research

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EDUCATION

University of Notre Dame

August 2013 - 2019

PhD

Department of Biological Sciences

Kenyon College

August 2008 - May 2012

Bachelor of Arts, Biology and Environmental Studies.

Magna cum laude

RESEARCH EXPERIENCE

ORAU Postdoctoral Researcher with USDA Forest Service

2023 - present

I am presently an Oak Ridge Associated Universities (ORAU) contractor working in the USDA Forest Service, Northern Research Station, Forest Inventory and Analysis (FIA) program in the the National Resource Assessment, Reporting, and Monitoring group. I am developing reproducible national frameworks to integrate FIA observations with tree ring data to improve estimation and driver attribution of annual forest carbon, growth, and mortality.

Postdoctoral Research Associate – University of Arizona

2019 - 2023

Laboratory of Tree Ring Research.

I generated forecasts of forest carbon stock and flux across the Interior West US and evaluated forecast uncertainties to inform forest management. In this work, I used Bayesian data assimilation approaches to combine two large data sets in a state space modeling framework: repeat censuses from the Forest Inventory and Analysis and annual tree ring growth records.

PhD Graduate Research Assistant – University of Notre Dame

2013 - 2019

My PhD explored how environmental changes affect both the large-scale distribution of forests, and treelevel responses to drought. Specifically, I quantified how historic changes in land-use, climate, and CO₂ impacted the stability and drought tolerance of the temperate savanna-forest biome boundary using historic spatial datasets, Bayesian modelling, dendrochronology, and stable isotope geochemistry.

Aquatic Research Technician – Cleveland Metroparks

2012 - 2013

Performed both fish and aquatic invertebrate sampling for restoration and management. Led ecology field trips, and contributed to outreach events.

Toucan Research Assistant – CATIE, Costa Rica

2013

Tracked range and seed dispersal of toucans to support the dissertation research of Dr. Landon Jones.

Kenyon Summer Science Research Scholar – Kenyon College

2011

Investigated the impacts of climate change on reproduction success of a threatened minnow species.

GRANTS AND FELLOWSHIPS

National Science Foundation Doctoral Dissertation Improvement Grant DEB-1701897.

2017

Center for Environmental Science Technology (CEST) Bayor Fellowship	2017
UNDERC Mentoring Fellowship	2017 2018
Arthur J. Schmitt Leadership Fellowship	2013-2019
Leslie Douglas Frye Scholarship to Attend Kenyon College	2008-2012
Dr. Kirby Walsh Sullivan Scholarship for Promising Students in Science	2009-2012

PUBLICATIONS

* indicates shared first authorship

Giebink, C.L.*, **Heilman, K.A.***, Cahoon, S.M.P., Domke, G.M. An assessment of the selection criteria, roles, and representativeness of indicator trees in the nationwide forest inventory of the United States. 2025. *Forest Ecology and Management*. <https://doi.org/10.1016/j.foreco.2025.123200>.

Evans, M.E.K., Adler, P.B., Angert, A.L., Dey, S.M.N., Girardin, M.P., **Heilman, K.A.**, Klesse, S., Perret, D.L., Sax, D.F., Sheth, S.N., Stenkovski, M. Williams, J.L. 2025. Reconsidering space-for-time substitution in climate change ecology. *Nature Climate Change*. <https://doi.org/10.1038/s41558-025-02392-0>

Domke, G.M.; Walters, B.F.; Smith, J.E.; Greenfield, E.J.; Giebink, C. L.; Ogle, S.M.; Steller, J.; Rewcastle, K.; Knott, J.A.; Coulston, John W.; **Heilman, K.A.**; Lang, A.K. Greenhouse gas emissions and removals from forest land, woodlands, urban trees, and harvested wood products in the United States, 1990-2022. 2024. *Resour. Bull. WO-102. Washington, DC: U.S. Department of Agriculture, Forest Service, Washington Office*. <https://doi.org/10.2737/WO-RB-102>.

Evans, M.E.* , Dey, Sharmila M.N.*,**Heilman, K.A.***, Tipton, J.R., DeRose, R.J., Klesse, S., Schulz, E.L., Shaw, J.D., Tree rings reveal the transient risk of extinction hidden inside climate envelope forecasts. 2024 *Proceedings of the National Academy of Sciences*. <https://doi.org/10.1073/pnas.2315700121>

Heilman, K.A., Dietze, M., Finley, A.O., Arizpe, A., Aragon, J., Gray, A., Shaw, J.D., Klesse, S., DeRose, R.J., Evans, M.E. Ecological forecasting of tree growth: regional fusion of tree-ring and forest inventory data to quantify drivers and characterize uncertainty. 2022 *Global Change Biology*. <https://doi.org/10.1111/gcb.16038>

Giebink, C.L., Domke, G.M., Fisher, R.A., Moore, D.J.P., **Heilman, K.A.**, DeRose, J.D., Evans, M.E.K. The policy and ecology of forest-based climate mitigation: challenges, needs, and opportunities. 2022. *Plant Soil*. <https://doi.org/10.1007/s11104-022-05315-6> 2022.

Novick, K., Jo, I., DOrangeville, L., Benson, M. Au, T.F., Barnes, M., Denham, S., Fei, S. **Heilman, K.A.**, Hwang, T., Keyser, T., Maxwell, J., Miniati, C., McLachlan, J.S., Pederson, N., Wang, L., Wood, J. D., P. Phillips, R.P. The Drought Response of Eastern US Oaks in the Context of Their Declining Abundance. 2022 *Bioscience*. <https://doi.org/10.1093/biosci/biab135>.

Heilman, K.A., Trouet, V., Pederson, N., Belmecheri, S., Berke, M.A., McLachlan, J.S., Increased water use efficiency leads to decreased precipitation sensitivity of tree growth, but is offset by high temperatures. 2021. *Oecologia*. <https://doi.org/10.1007/s00442-021-04892-0>

Evans, M.K., DeRose, R. J., Klesse, S., Girardin, M. P., **Heilman, K.A.**, Alexander, M.R., Arsenault, A., Babst, F., Bouchard, M., Cahoon, S. M. P., Campbell, E., Dietze, M.C., Duchesne, L., Frank, D.C., Giebink, C.L., Gomez-Guerrero, A., Gutierrez Garca, G., Villela Gaytn, S.A., Edward H. Hogg, E.H., Metsaranta, J., Ols, C., Rayback, S. A., Reid, A., Ricker, M., Schaberg, P. G., Shaw, J. D., Sullivan, P. F. The case for adding tree rings to North America's national forest inventories: an essential tool to guide drawdown of atmospheric CO₂. 2021. *Bioscience*. <https://doi.org/10.1093/biosci/biab119>

Walker, A. P.; Kauwe, M. G. D.; Bastos, A.; Belmecheri, S.; Georgiou, K.; Keeling, R. F.; McMahon, S. M.; Medlyn, B. E.; Moore, D. J. P.; Norby, R. J.; Zaehle, S.; AndersonTeixeira, K. J.; Battipaglia, G.; Brienen, R. J. W.; Cabugao, K. G.; Cailleret, M.; Campbell, E.; Canadell, J. G.; Ciais, P.; Craig, M. E.; Ellsworth, D. S.; Farquhar, G. D.; Fatichi, S.; Fisher, J. B.; Frank, D. C.; Graven, H.; Gu, L.; Haverd, V.; **Heilman, K.A.**; Heimann, M.; Hungate, B. A.; Iversen, C. M.; Joos, F.; Jiang, M.; Keenan, T. F.; Knauer, J.; Krner, C.; Leshyk, V. O.; Leuzinger, S.; Liu, Y.; MacBean, N.; Malhi, Y.; McVicar, T. R.; Penuelas, J.; Pongratz, J.; Powell, A. S.; Riutta, T.; Sabot, M. E. B.; Schleucher, J.; Sitch, S.; Smith, W. K.; Sulman, B.; Taylor, B.; Terrer, C.; Torn, M. S.; Treseder, K. K.; Trugman, A. T.; Trumbore, S. E.; Mantgem, P. J. van; Voelker, S. L.; Whelan, M. E.; Zuidema, P. A. 2020. Integrating the Evidence for a Terrestrial Carbon Sink Caused by Increasing Atmospheric CO₂. 2020. *New Phytologist*. <https://doi.org/10.1111/nph.16866>.

Broderick, C.M., **Heilman, K.A.**, T.A. Patterson, J.A. Peters, J.S. McLachlan. Sharp Savanna-forest Transitions in the Midwest Followed Environmental Gradients but are Absent from the Modern Landscape. 2018. *The American Midland Naturalist*, 180(1):1-17. <https://doi.org/10.1674/0003-0031-180.1.1>

MANUSCRIPTS IN PREP/SUBMITTED

Shuman, I.; McLachlan, J.S.; Peters, J.A.; **Heilman, K.A.** Kowalski, C.; Willson, A., Both environmental gradients and feedbacks determine the historical distribution of forest, savanna, and prairie ecosystems in the Midwestern United States. In Review.

Heilman, K.A., Staver, A.C., Williams, J.W. Goring, S., Paciorek, C.J., Tipton, J.R., McLachlan, J.S. The anthropogenic collapse of savanna and forest alternative stable states in the US Midwest. In Revision.

PRESENTATIONS

Invited Presentations

Heilman, K.A., Dawson, A., Dietze, M.C., DeRose, R.J., Shaw, J.D, Evans, M.E.K., Domke, G.M. Linking tree ring and forest inventory data: towards annual, climate sensitive estimates of tree growth and aboveground biomass. FIA Science Symposium, Virtual, November 20, 2024.

Heilman, K.A., Dietze, M.C., Knott, J., Walters, B., Domke, G.M. Growth reductions are an early warning sign of tree mortality in Northeastern US Detectable in Historic Forest Inventories. FIA Science Symposium, Virtual, November 19, 2024.

Heilman, K.A., Dawson, A., Dietze, M.C., DeRose, R.J., Shaw, J.D, Knott, J., Walters, B., Domke, G.M., Evans, M.E.K. Incorporating ecological complexity in growth and mortality models through hierarchical data fusion of tree ring and forest inventory data. Ecological Society of America, Long Beach,

CA, August 6, 2024.

Kelly Heilman. Assessing drivers of tree growth and mortality with forest inventory and tree ring data. Honduras ICF and USFS Science Exchanges January 23, 2024.

Heilman, K.A., DeRose, R.J., Shaw, J.D., Finley, A.O., Dietze, M.C., Evans, M.E.K. Ecological forecasting of forest biomass with tree-ring and forest inventory networks. Growth and Yield Innovations Conference 2023, Canmore, Alberta, CA. June 20, 2023.

Heilman, K.A., DeRose, R.J., Shaw, J.D., Finley, A.O., Dietze, M.C., Klesse, S., Evans, M.E.K. Bayesian fusion of tree ring and national forest inventory data to improve forecasts of forest carbon. University of Arizona Quantitative Biology Colloquium, Tucson, Arizona, April 4th, 2023.

Heilman, K.A., DeRose, R.J., Shaw, J.D., Finley, A.O., Dietze, M.C., Aragon, J., Grey, A., Arizpe, A., Klesse, S., Evans, M.E.K. Ecological forecasting of ponderosa pine forest biomass in the Interior West US. American Geophysical Union, Chicago/Virtual, December 16th, 2022

Heilman, K.A., DeRose, R.J., Shaw, J.D., Finley, A.O., Dietze, M.C., Aragon, J., Grey, A., Arizpe, A., Klesse, S., Evans, M.E.K. Fusing tree ring and national forest inventory data to improve forecasts of forest carbon. FIA Stakeholder Science Meeting, Virtual, October 14th, 2022.

Kelly Heilman. Ecological Forecasting of Ponderosa Pine Forest Carbon in the Inter-mountain West. MtnClim Conference Early Career Session, Rocky Mountain Biological Laboratory, Gothic, CO, September 15th, 2022

Kelly Heilman. Quantifying uncertainty around forest carbon storage and uptake: a Bayesian state-space model approach. Los Alamos National Laboratory, New Mexico, Presented Virtually, August 16th, 2022

Kelly Heilman. Listening to the trees: How tree rings help us predict future forests. Science on Screen, The Loft Cinema, sponsored by the Sloan foundation and The Center For Creative Photography, April 19, 2022

Kelly Heilman and Justin DeRose. Fusing tree ring growth and national forest inventory data to forecast tree growth and aboveground biomass across scales. Utah State University Wildland Seminar, March 24, 2022

Kelly Heilman. Quantifying uncertainty around future forest carbon storage and uptake: a Bayesian state-space model approach RTG Showcase, University of Arizona, March 1, 2022

Heilman, K.A., DeRose, R.J., Shaw, J.D., Finley, A.O., Dietze, M.C., Aragon, J., Grey, A., Arizpe, A., Klesse, S., Evans, M.E.K. Fusing tree ring and forest inventory data for ecological forecasting of tree growth responses to climate change. American Association of Geographers Annual meeting, Frontiers in Dendrochronology Session. April, 2021

Kelly Heilman. Assimilation of tree ring and forest inventory data to forecast future growth responses of *Pinus ponderosa*. August, 2020. Laboratory of Tree Ring Research Department Seminar. October 14th, 2020

Heilman, K.A., DeRose, R.J., Shaw, J.D., Finley, A.O., Dietze, M.C. and Evans, M.E.K. Assimilation of Tree Ring and Forest Inventory Data to Constrain Uncertainty in Ecological Forecasts. SIAM Math-

ematics of Planet Earth, Virtual. August 14, 2020

Kelly Heilman. Forecasting long-term effects of environmental changes on vegetation: from trees to biomes. Kenyon College Biology Department Seminar. Gambier, OH, November 14, 2019

Kelly Heilman. Vegetation Responses to Long-term Environmental Changes at a Savanna Forest Boundary. Laboratory of Tree Ring Research Department Seminar. University of Arizona, October 9, 2019

Heilman, K.A., McLachlan, J.S., Pederson, N., Trouet, V., Belmecheri, S. Tree growth in a changing world: Factors mediating the effects of CO₂ on tree ring growth and Water Use Efficiency. Ecological Society of America. New Orleans, LA, August 8, 2018

Kelly Heilman. Tree growth in a changing world: Oak drought sensitivity trends at the savanna-forest boundary. Morton Arboretum, Chicago, IL. May 21, 2018

Heilman, K.A., McLachlan, J.S. Quantifying vegetation structure and environmental drivers across a historic Prairie-Forest Biome Boundary. January 8, 2016. Harvard Forest Seminar Talk.

Contributed Oral Presentations

Kelly Heilman, Andria Dawson, Michael C. Dietze, Margaret E. K. Evans, Grant M. Domke. Toward a tree-ring enhanced National Forest Inventory: Data assimilation approaches to estimate inter-annual variability in forest carbon and drivers. American Geophysical Union. December 15th, 2025.

Kelly Heilman, Andria Dawson, Michael C. Dietze, Eric Scott, Dani Steinberg, Brian F. Walters, Margaret E. K. Evans, Grant M. Domke. Tree-ring enhanced forest inventory estimates to better characterize variability in carbon stocks for Greenhouse Gas Inventory reporting. 11th World Dendro Conference, Livingstone, Zambia. July 28th, 2025.

Kelly Heilman, Michael C. Dietze, Jonathan Knott, Brian Walters, Grant M. Domke. Growth reductions are an early warning sign of tree mortality in Northeastern US Detectable in Historic Forest Inventories. American Geophysical Union, Washington D.C., December 11, 2024.

Kelly Heilman, Andria Dawson, R. Justin DeRose, John D. Shaw, Michael Dietze, Grant M. Domke, Margaret E.K. Evans. Using ecological forecasting to quantify sensitivity of forest carbon to tree growth and mortality processes. Ecological Forecasting Initiative Conference 2024, Helsinki, Finland. June 12, 2024. Presented Virtually

Kelly A. Heilman, Michael Dietze, Andrew O. Finley, John D. Shaw, R. Justin DeRose, Andria Dawson, Grant M. Domke, Margaret E.K. Evans. Lightning talk: Climate-sensitive forest C forecasts from tree rings and forest inventory data indicate Ponderosa pine forests of the interior west will be a net source of forest C. NSF Macrosystems PI Meeting February 7, 2024.

Kelly A. Heilman, Michael Dietze, Andrew O. Finley, John D. Shaw, Stefan Klesse, R. Justin DeRose, Margaret E.K. Evans. Symposium talk: Fusing tree-ring growth and national forest inventory data to forecast tree growth and aboveground biomass across scales. Ameridendro, Montreal, CA. June 27th 2022.

Kelly Heilman. Ecological forecasting of tree growth and stand-level forest carbon storage and uptake. Ecological Forecasting Initiative Virtual Conference, May 23rd, 2022

- Kelly Heilman. Ecological forecasting of tree growth and stand-level forest carbon storage and uptake. Ecological Forecasting Initiative Virtual Conference, May 23rd, 2022
- Heilman, K.A., Dietze, M.C., Finley, A.O., Arizpe, A., Aragon, J., Gray, A., Shaw, J.D., Klesse, S., DeRose, R.J., Evans, M.E.K. Carbon accounting of ponderosa pine forests across the interior western U. S. based on tree-ring and forest inventory data: Drivers of carbon stock and flux and their uncertainties. August, 2021. Ecological Society of America
- Heilman, K.A., DeRose, R.J., Shaw, J.D., Finley, A.O., Dietze, M.C., and Evans, M.E.K.. Assimilation of tree ring and forest inventory data to forecast future growth responses of *Pinus ponderosa*. August, 2020. Ecological Society of America
- Heilman, K.A., McLachlan, J.S., Staver, A.S., Paciorek, C., Tipton, J., Williams, J.W.. Loss of alternative savanna and forest stable states due to historic and modern land-use. August, 2019. Ecological Society of America
- Heilman, K.A., McLachlan, J.S., Pederson, N., Trouet, V. Belmecheri, S.. Anthropogenic changes impact drought sensitivity across a savanna-forest biome boundary. 10th World Dendro Conference. Thimphu, Bhutan. June 2018
- Heilman, K.A., McLachlan, J.S., Drought sensitivity changes over the last century at the North American savanna-forest boundary. American Geophysical Union. December 11th, 2017
- Heilman, K.A., McLachlan, J.S., Staver, A.C., Stabilization of savanna-forest alternative states to closed forests after 150 years of land use and climate changes. August, 2017 Ecological Society of America
- Heilman, K.A., McLachlan, J.S.. What controls temperate savanna tree growth? July, 2016. North American Prairie Conference.
- Heilman, K.A., McLachlan, J.S. Using GIS to investigate controls on vegetation at the prairie-forest boundary. November 16th, 2016. University of Notre Dame GIS day.
- Heilman, K.A., McLachlan, J.S., Staver, A.C.. Investigating controls of tree density and resilience to environmental changes at the prairie-forest boundary. November 2016. Biofrass Seminar Presentation, University of Notre Dame.
- Poster Presentations**
- Heilman, K., Dawson, A., Dietze, M.C., Evans, M.E.K., and Domke, G.M., Bayesian Fusion of Tree Ring Data and Forest Inventory Data to Improve Forecasts of Tree Growth, C Sequestration Variability, and Climate Sensitivity. Ecological Forecasting Initiative Conference, May, 2025.
- Heilman, K., Dawson, A., Giebink, C.L., Dietze, M.C., Evans, M.E.K., and Domke, G.M., Fusion of tree ring data and forest inventory data to improve estimates of tree growth, C sequestration variability, and climate sensitivity. American Geophysical Union, December, 2024.
- Heilman, K., DeRose, R.J., Shaw, J.D., Finley, A.O., Dietze, M.C., Domke, G.M., and Evans, M.E.K. Challenges and opportunities of using forest inventory and tree ring data for forest C accounting. American Geophysical Union, December, 2023.

Heilman, K., DeRose, R.J., Shaw, J.D., Finely, A.O., Dietze, M.C., and Evans, M.E.K. Carbon accounting of ponderosa pine forests across the interior western US based on tree ring and forest inventory data. American Geophysical Union, December, 2021.

Heilman, K., DeRose, R.J., Shaw, J.D., Finely, A.O., Dietze, M.C., and Evans, M.E.K.. Fusion of tree-ring and forest inventory data to quantify C cycle drivers and characterize forecast uncertainty. North American Carbon Project. Virtual meeting March, 2021.

Heilman, K.A., Berke, M., Pederson, N., Trouet, V., Belmecheri, S., McLachlan, J.S.. Impacts of anthropogenic changes on water use efficiency and climate sensitivity: A data-model comparison. American Geophysical Union 2018.

Heilman, K.A., McLachlan, J.S., Staver, A.C., The role of disappeared disturbances in driving the North American prairie-forest boundary. American Geophysical Union, December 2016.

Heilman, K., McLachlan, J. Feng, X. Convergent Ecosystems from Divergent Environmental Drivers: revisiting drivers of the past prairie-forest ecotone across the North American Prairie Peninsula. American Geophysical Union, December, 2015.

Heilman, K., Pitcher, T.P., Heithaus, E.R. Is sperm motility correlated with red spawning coloration in Redside Dace (*Clinostomus elongatus*)? Kenyon College Summer Science Scholar Research Symposium 2011

Co-Authored Presentations

Ruby Celeste Morillo, Kelly A. Heilman, Alyssa M. Wilson, Andria Dawson, Impacts of climate on stand and species-level on aboveground biomass increment estimates. World Dendro 2025.

Andria Dawson, Ruby Celeste Morillo., Alyssa M. Wilson, Kelly A. Heilman, Estimating aboveground biomass from census and tree ring data. World Dendro 2025.

Danielle Steinberg, Eric Scott, Renata Diaz, Kelly Heilman, Brian F Walters, Grant M Domke, and Margaret Evans. Threading the Needle through Multi-year Forest Inventory Intervals: Advancing Forest Carbon Estimation with forestTIME Annualized Interpolation. AGU 2025.

Cameron Webb, Michael Dietze, Dongchen Zhang, Margaret Evans and Kelly Heilman, Using National Forest Inventory Data to Advance Carbon Cycle Model-Data Assimilation. AGU 2025.

Grant M Domke, Brian F Walters, Colin M Beier, Christopher Edgar, Margaret E.K. Evans, Songlin Fei, Andrew O Finley, Kelly Heilman, Jonathan Knott, Jeremy W Lichstein, Adam R Martin, Sassan S Saatchi, Toward integration of forest inventories, remotely sensed information, and models to characterize carbon stocks and the demographic drivers of change on forest land in the United States. AGU 2024.

Margaret Evans, Kelly Heilman, Andria Dawson, Justin DeRose, Renata Diaz, Michael Dietze, Grant Domke, John Shaw, Danielle Steinberg, Annual-resolution Forest Carbon Estimation and Forecasting Based on Forest Inventory and Tree-ring Data. AGU 2024.

Cazimir Kowalski, Luca Menicali, Stefano Castruccio, Kelly Heilman³ and Jason S McLachlan, Modeling Sharp Vegetation Boundaries from Continuous Environmental Space, Species Covariance, and Spatial Processes. AGU 2024.

Jonathan Knott, Grant Domke, Brian F Walters, Stephen M Ogle, Shannon Spencer, Courtney Giebink and Kelly Heilman. An Inventory-based Land Use and Land Use Change Classification for the United States

Daniel Moreno-Fernández, Kelly A. Heilman, Ana Aguirre, Iciar Alberdi, Isabel Caellas, Sonia Conds, Margaret E.K. Evans, Isabel Dorado-Lin. TRACE, May 2023.

Evans, M.E.K, DeRose, R.J., Klesse, S., Girardin, M.P., Heilman, K.A., Dietze, M.C., Shaw, J.D., Babst, F., Cahoon, S.M.P., Duchesne, L., Frank, D., Giebink, C.L., Martinez Gutierrez, G., Huntingdon Hogg, E., Metsaranta, J., Rayback, S.A., Schaberg, P.G., Alexander, M.R., Sullivan, P. and Arsenault, A. Enhanced Forest Ecosystem Monitoring for Carbon Accounting and Climate Mitigation: the Case for Adding Tree Rings to North American's National Forest Inventories. AGU 2021.

TEACHING AND PROFESSIONAL EXPERIENCE

World Dendro Workshop: Forest growth and carbon with uncertainty from tree-rings and census data *July 27th, 2025*

Livingstone, Zambia

Co-organized a one-day workshop with 15 participants on combining tree ring data with forest census data. In this workshop, I co-created and led a tutorial that demonstrates the value of combining tree ring and census data to estimate forest growth, biomass, and forest Carbon using a Bayesian state-space modeling approach. We developed a [workshop website](#), using [github to share code](#), data, and results examples, which provided both a "no-code," and an R code opportunity to engage with the material, to reduce barriers to learning these methods, and increase access.

Dendroecology and State-Space model workshop *Nov. 2023*

University of Arizona

Co-organized a three day workshop with 40 participants on combining tree ring data with forest inventory data. In this workshop, I created and led a code tutorial for fitting Bayesian state-space models of tree growth, ([publicly available on github](#)), presented a research seminar, and worked with participants to include their data into similar analyses.

Conservation Biology 406/506 - Instructor *Spring 2022*

University of Arizona

Instructor of record for Conservation Biology, a 400-level undergraduate course, and 500-level graduate course cross listed in Ecology and Evolutionary Biology, and School of Natural Resources and the Environment

Guest lecture in DISC (Dendrochronology Intensive Summer Course)

University of Arizona, Laboratory of Tree Ring Research – June 2022

Invited guest lecture on ecological forecasting and uncertainty quantification in dendroecology.

Guest lecture in Forest Ecology Class

University of Arizona, Laboratory of Tree Ring Research – Fall 2021

Guest lectured and led discussion on the topic of forest carbon fertilization.

Guest lecture in Graduate Seminar on Ecological Forecasting

University of Arizona, Laboratory of Tree Ring Research – Fall 2019

Guest lectured and led discussion on a Bayesian State-Space model, Intro to Bayesian Statistics, and led discussion on the role of Ecological Forecasting in our own research.

Workshop on Introduction to Bayesian Statistics in R

University of Arizona Research Bazaar – May 2020 and May 2021

Developed and taught a workshop on an introduction to Bayesian Statistics and how to implement simple Bayesian models in R. All course materials are available on github.

General Ecology Lab Lead Teaching Assistant

2018

University of Notre Dame

Contributed to ongoing course development by creating new lesson plans and leading writing workshops for students. I also assisted the Teaching Assistants with lessons, leading field trips, and developing student writing skills.

General Ecology Lab Teaching Assistant

2014, 2016, 2017

University of Notre Dame

Prepared fieldwork experiments and discussions, lead field trips, advised students on data analysis, developed student scientific writing and intellectual virtues, and graded all material.

UNDERC Summer Course Research Mentor

2017-2018

Notre Dame Environmental Research Center

Mentored five undergraduate students on independent summer research projects. This included teaching the students dendrochronology methods, forestry field data collection, implementing statistics using R, and the scientific writing process.

General Biology Lab - Teaching Assistant

2013

University of Notre Dame

Prepared introductory lectures, set up lab equipment, advised students on independent research projects, and graded all assignments.

Aquatic Systems Biology - Laboratory Teaching Assistant

2011

Kenyon College

Teaching duties included organizing lab materials, bringing students on field expeditions, and teaching students aquatic ecology methods.

Introductory Biology Lab - Teaching Assistant

2010-2012

Kenyon College

Assisted instructor in preparing lecture and experimental design, and led students through experimental protocols.

Introductory Biology Lab and Microscopy Lab Preparation Assistant

2010-2011

Kenyon College

Prepared lab materials for each lab meeting, and helped to develop lesson plans each week.

VOLUNTEER AND SERVICE

Organizational

Tucson Bird Alliance**May 2026***Board of Directors Member*

I recently joined the Board of Directors at Tucson Bird Alliance (previously Tucson Audubon Society), where I volunteer as a part of the the Conservation Action Committee.

Ecological Forecasting Initiative Conference Postdoctoral Advising Member**2022****Tucson-based task force of Arizona Science Policy Network****2021-present***co-organizer and member*

Starting up a Tucson-based task force of the Arizona Science Policy Network, focused on advocating for science-based solutions to issues facing our communities.

Jogging Organization for Graduate Students (JOGs)**2018-2019***President*

Co-founded a social running club for graduate students to network with fellow runners and researchers.

University of Notre Dame Graduate Student Union**2017-2018***Social chair and community engagement chair*

Coordinated social and community engagement events for graduate students and families, including a Charity Gala. Served as a voting member of Notre Dame Graduate Student Union.

Society of Schmitt Fellows University of Notre Dame**2015-2017***Treasurer*

Handled Society of Schmitt Fellows annual budget and co-organized a fund for conference presentation travel grants.

Social Chair

Co-organised monthly social activities for STEM graduate students.

Outreach Committee

Planned outreach events in the community.

Reviewer

Journals: Science, Ecosphere, Dendrochronologia, Scientific Reports, Forest Ecology and Management, Michigan Academician.

Proposals: National Science Foundation Biology Program, Joint Fire Science Program

COMMUNITY OUTREACH AND ENGAGEMENT

International Programs**Science Exchange with Honduras Instituto de Conservación Forestal and USDA Forest Service****January 2024**

Participated in discussions about capabilities of National Forest Inventory data in Honduras, sampling strategies, and the potential roles of tree ring data and National Forest Inventory data in meeting the challenges of climate, disturbances, and land-use changes.

Community Engagement

Reframing Wildfire

March 2, 2023

Co-lead and co-organized a community panel discussion featuring artists, scientists, local conservation organizations and the US Forest Service to discuss how we can meaningfully (re)frame the wildfire and human interactions with the landscape. Acquired funding and support from the American Geophysical Union, Laboratory of Tree Ring Research, and Crooked Tooth Brewing Company.

Science on Screen at the Loft Cinema

April 19, 2022

Presented my research as an introduction to the film "The Hidden Life of Trees" to ~300 attendees. This was a part of the Loft's Earth day film festival, and Science on Screen is sponsored by the Sloan Foundation and the Center for Creative Photography.

Pine Chant: Collaboration with musicians and artists

2021

Contributed to an art-science collaboration between the UArizona School of Music, School of Art, the Center for Creative Photography, and the Laboratory of Tree-Ring Research. In this collaboration, tree ring data from National Forest Inventory plots were used as inspirations for a piece composed by Lachlan Skipworth, "Pine Chant", ([Available on Spotify](#)), and was performed by three wind instruments at the Laboratory of Tree Ring Research, Center for Creative Photography, Arizona-Sonora Desert Museum, University of Arizona School of Music, and the 2022 Conference of the International Double Reed Society. It was made possible by a University of Arizona Production Faculty Seed Grant and the College of Fine Arts Fund for Excellence. I contributed by discussing our research, providing tree ring data used to compose the piece, contributing the scientific interpretation of the data and generating simple visualizations (videos, charts). [Watch a recording and see more information on Sara Fraker's website.](#)

Macrosystems exhibit for Laboratory of Tree Ring Research lobby

2020

Helped to develop and create a series of outreach videos about our research forecasting forest responses to climate change, which is a permanent video exhibit in the Laboratory of Tree Ring Research lobby.

Southern Arizona Foresters

2019

Presented a research talk and led a discussion during the Southern Arizona Foresters Luncheon.

STEM Programming at the Boys and Girls Club of South Bend

2016-2017

Organized weekly science activities for elementary school students in the Boys and Girls Club after school program.

Planting Science Mentor

2017-2018

Mentored High School Science classes during independent research projects online.

Volleyball Head Coach

2017

Coached 6th grade volleyball team at St. Pius X School in Granger, IN.

Science Policy

Memo on the Filibuster on behalf of Union of Concerned Scientists

July 2021

Co-wrote a memo on the historical use of the filibuster, its misuses, and details potential outcomes of filibuster reform, as it relates to Senator Mark Kelly's policy objectives. This was a joint effort between members of the Arizona Science policy network, and was sent on behalf of the Union of Concerned Scientists.

AGU Broadening participation in STEM advocacy day

June 2021

Met with staffers from Raul Grijavla, Kirsten Sinema, and Senator Mark Kelly's office to ask for them to support several bills that would help broaden and diversify participation in STEM, including the RESTART STEM Act (S. 1297), the STEM Opportunities Act, and the Reconciliation in Place Names Act. This meeting was on behalf of American Geophysical Union.

AGU Voices for Science Advocacy Day

May 2021

Met with staffers from Raul Grijavla, and Senator Mark Kelly's office to ask for them to support increases in science funding for NSF and USGS in the FY 22 budget request. This meeting was on behalf of American Geophysical Union and the Voices for Science Program.

Meeting with policymakers to strengthen the role of science in democracy

April 2021

Met with Coral Evans (former Mayor of Flagstaff, and current staffer for Senator Mark Kelly) to discuss science policy, voting rights, and democracy. We thanked Sen. Kelly for his support of the For the People act, and spoke in support of the John Lewis VRA and the Environmental Justice for All Act. This meeting was on behalf of Union of Concerned Scientists.

Arizona Science Policy Day

March 2021

Led and organized a discussion of nature-based climate solutions to address the climate crisis, developed a one-pager to disseminate to Arizona Legislators, and engage with students interested in climate-related policies. This event was a part of Arizona Science Policy Networks Science Day

Organized meetings with policymakers for Climate Advocacy Week

October 2020

Led and organized a meeting with our Arizona Senators' offices (Kirsten Sinema[D], Martha McSally[R]) and our local House representative (Ann Kirkpatrick[D]) to discuss policy to address the climate crisis and offer our expertise as climate and planetary scientists. This event was a part of AGU's Climate Advocacy Week

HONORS, AWARDS, AND CERTIFICATES

Ecological Forecasting Award from the Ecological Society of Americas Statistical Ecology section *2022*

CIRTL Postdoc Pathway Fellows Program Certificate (Associate Level I and Practitioner Level II) *2022*

American Geophysical Union Voices for Science Fellow *2021-2022*

Postdoctoral Professional Development Certificate *2020*

University of Arizona Postdoc Speed Talk Finalist *2020*

Saint Joseph Academy Alumni 25 under 35 Award *2016*

Distinction Award Biology Senior Exercise *2012*

TEACHING AND COURSE DEVELOPMENT TRAINING

Center for Integration of Teaching and Research Learning

University of Arizona – Fall 2021

Earned the the CIRTL Postdoc Pathway Fellows Program practitioner certificate, upon successful completion of an intensive one-year higher education teaching preparation program. This program involves a selective application process, a course on evidence-based STEM teaching, biweekly faculty learning community discussions, teaching workshops, and a collaborative co-teaching project.

Teaching Where the Magic Happens: Field Course Design Workshop

Ecological Society of America – August 2017

Attended a workshop to develop short field courses, discussed risk management, best teaching practices, and dealing with logistics while developing remote field-based learning experiences.

A Practical Guide to Teaching and Learning in STEM

University of Notre Dame Kaneb Center — January 2018

Participated in a workshop lead by Rebecca Brent and Richard Felder, focused on designing successful sciences courses with measurable learning objectives, active learning, and appropriate assessment.

Striving for Excellence in Teaching Certificate

University of Notre Dame Kaneb Center — March 2018

Participated in 5+ workshops on teaching and course design at University of Notre Dame, and provided reflections on how the workshops provided skills to improve my own teaching.

Wakonse Conference on College Teaching

May 24-29th 2018

Participated in the 2018 Wakonse Teaching Fellowship Conference in Michigan, where college educators share teaching methods, reflect on personal strengths and weaknesses as college teachers, and craft short- and long-term professional goals.

Introduction to Course Design University of Notre Dame Kaneb Center — Spring 2017

Participated in a 5-week workshop series to improve my course design and help develop teaching philosophy.

GET-STEM Program

University of Notre Dame Kaneb Center — Fall 2016

Volunteered in the Graduate Educators Teaching Science Technology Engineering and Math program.

OPPORTUNITIES CANCELED DUE COVID-19 AND OTHER DISASTERS

In addition to most of the conferences, seminars, and research collaborations moved online or delayed during my post-doctoral research, several engagement opportunities were cancelled, or otherwise impeded by the ongoing COVID-19 pandemic, and other disasters. I list these here:

Presented virtually at the Ecological Forecasting Initiative Conference 2024 due to COVID illness.

Invited Talk at the Forest Growth and Yield Innovations Conference 2022 in Canmore, Alberta, Canada. Conference was cancelled and rescheduled to 2023.

Invited Talk at Los Alamos Math Days at Los Alamos National Laboratory (May 2022) Conference rescheduled due to wildfire and evacuation notices near Los Alamos National Laboratory. I presented virtually due to illness.

Co-Instructor of Using Bayesian statistics in Dendroecology at North American Dendro-Ecological Fieldweek (NADEF), July 2020

Science Cafe Speaker at Tucson Festival of Books, March 2020